Endocrinology

Z352

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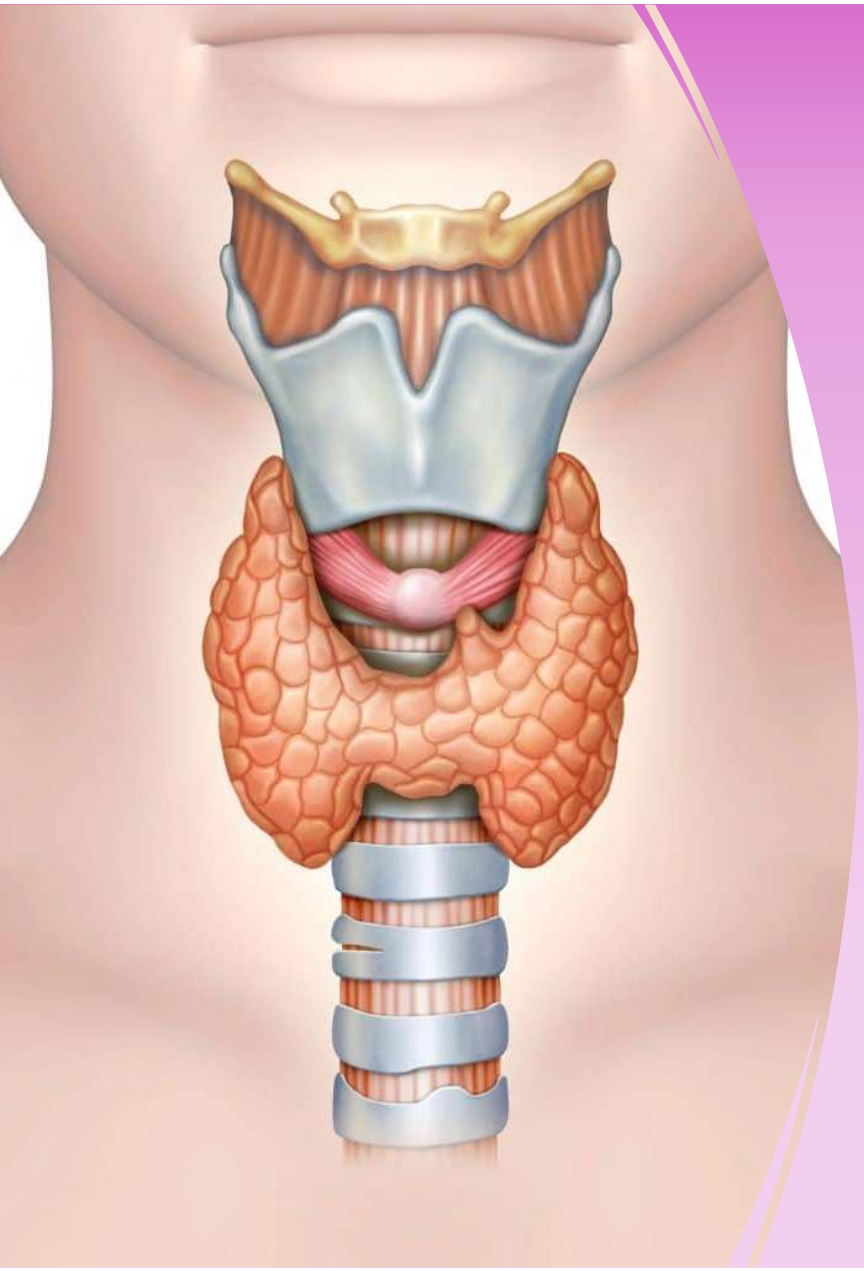


Thyroid and parathyroid glands

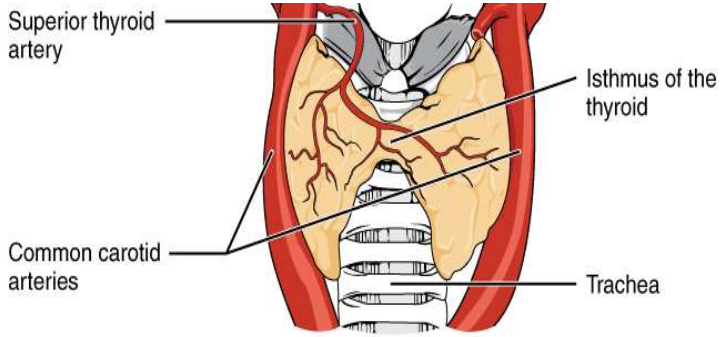
Thyroid gland

- The **thyroid gland,** a butterfly-shaped organ**,** is a large endocrine gland located in the neck. It is attached to the trachea just below the larynx.

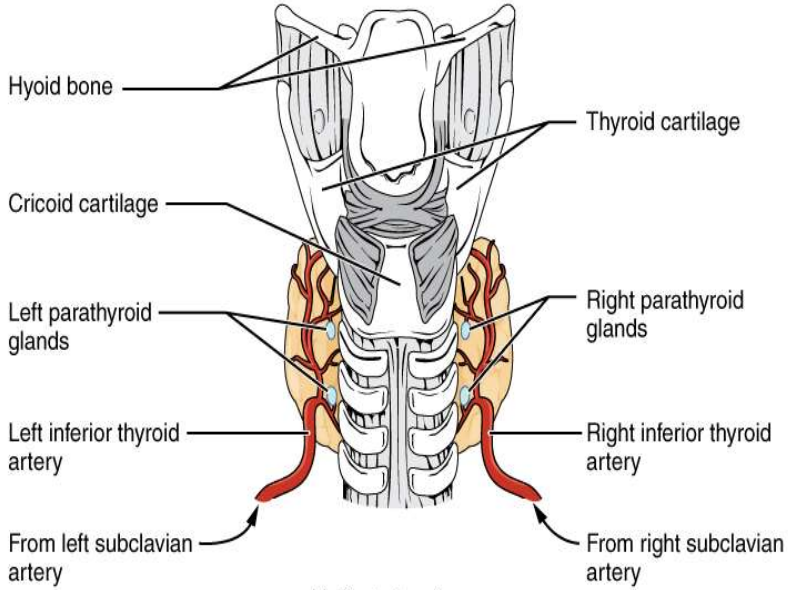
- The medial region, called the **isthmus**, is flanked by wing-shaped left and right lobes.

- Each of the thyroid lobes is embedded with parathyroid glands, primarily on their posterior surfaces.

- The size of the gland is quite variable, depending on heredity, environment, and nutritional factors.



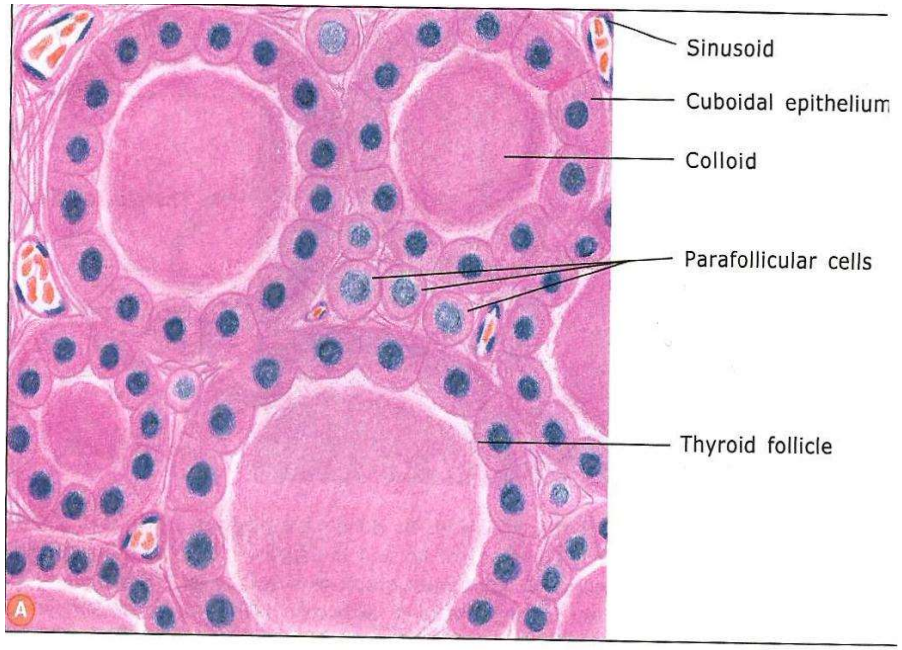
a) Anterior view



b) Posterior view

Anatomy of thyroid gland

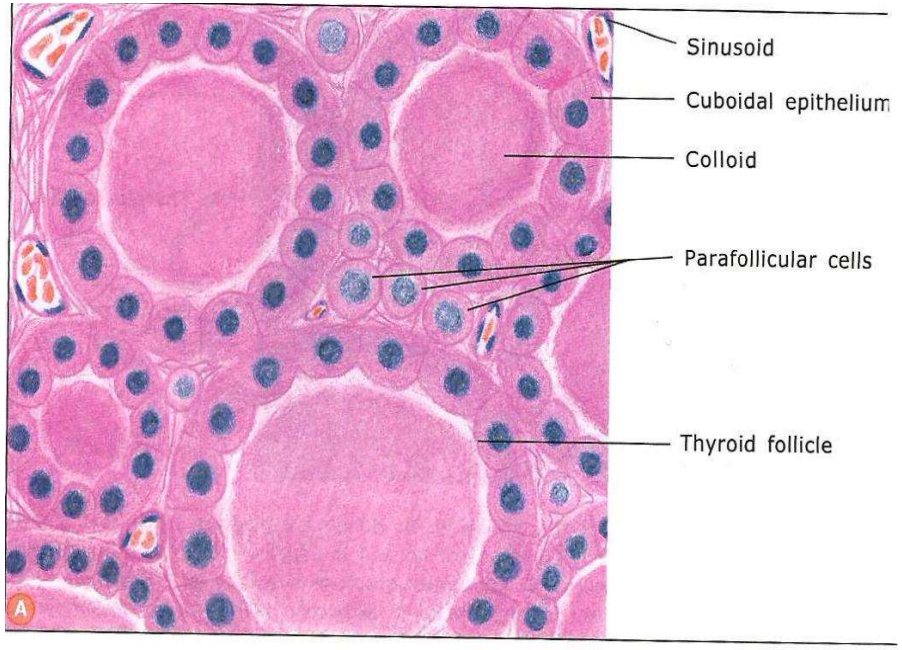
- The thyroid gland is primarily composed of thyroid follicles.
- Each follicle contains a central cavity filled with a sticky fluid known as a colloidal storage protein called thyroglobulin.
- The wall of the follicle is made up of cuboidal epithelial cells.
- The colloid is the center of thyroid hormone production, and that production is dependent on the hormones' essential and unique component: iodine.



Normal thyroid seen microscopically consists of follicles lined by cuboidal epithelium and filled with pink, homogenous colloid.

Anatomy of thyroid gland

- Lying in between the thyroid follicles are groups of separate larger epithelial cells, the *parafollicular C cells,* responsible for the secretion of a peptide hormone **calcitonin,** involved in the control of calcium metabolism.



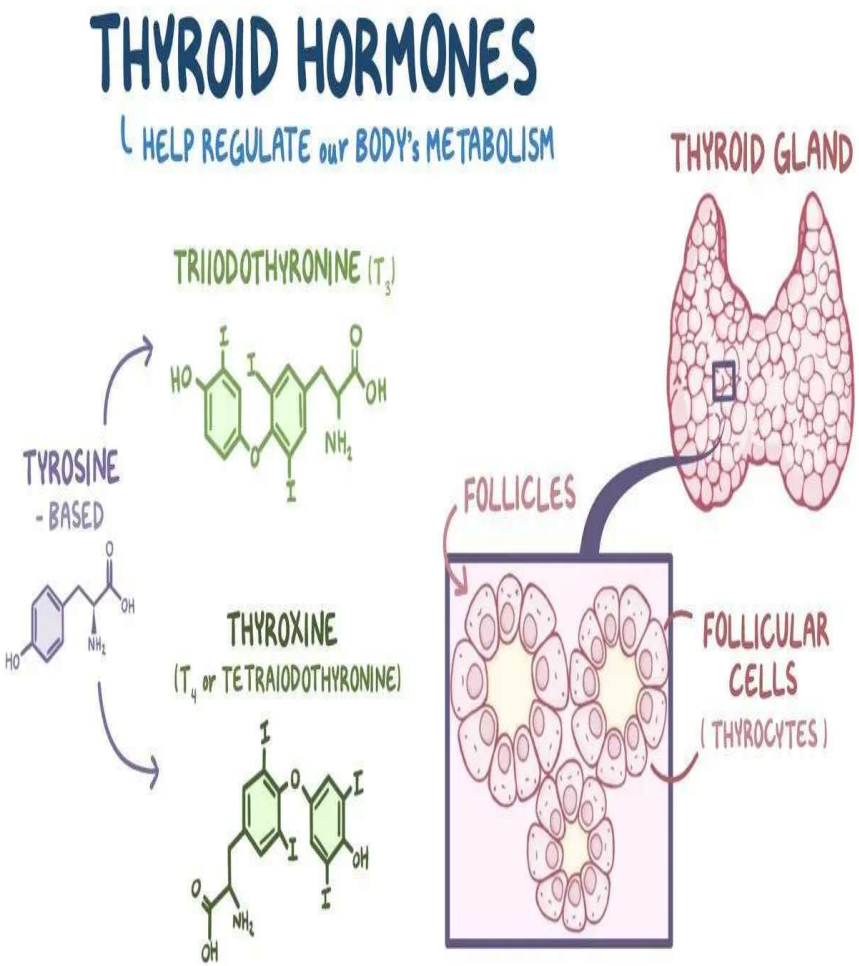
Thyroid Hormones

- The two active hormones secreted by the thyroid are iodinated derivatives of the amino acid *tyrosine;* these are **thyroxine** (**T4**; about 90% of output) containing four iodine atoms, and **triiodothyronine** (**T3**; about 10% of output) containing three iodine atoms.

- The term **'thyroid hormone'** thus encompasses both T3 and T4.

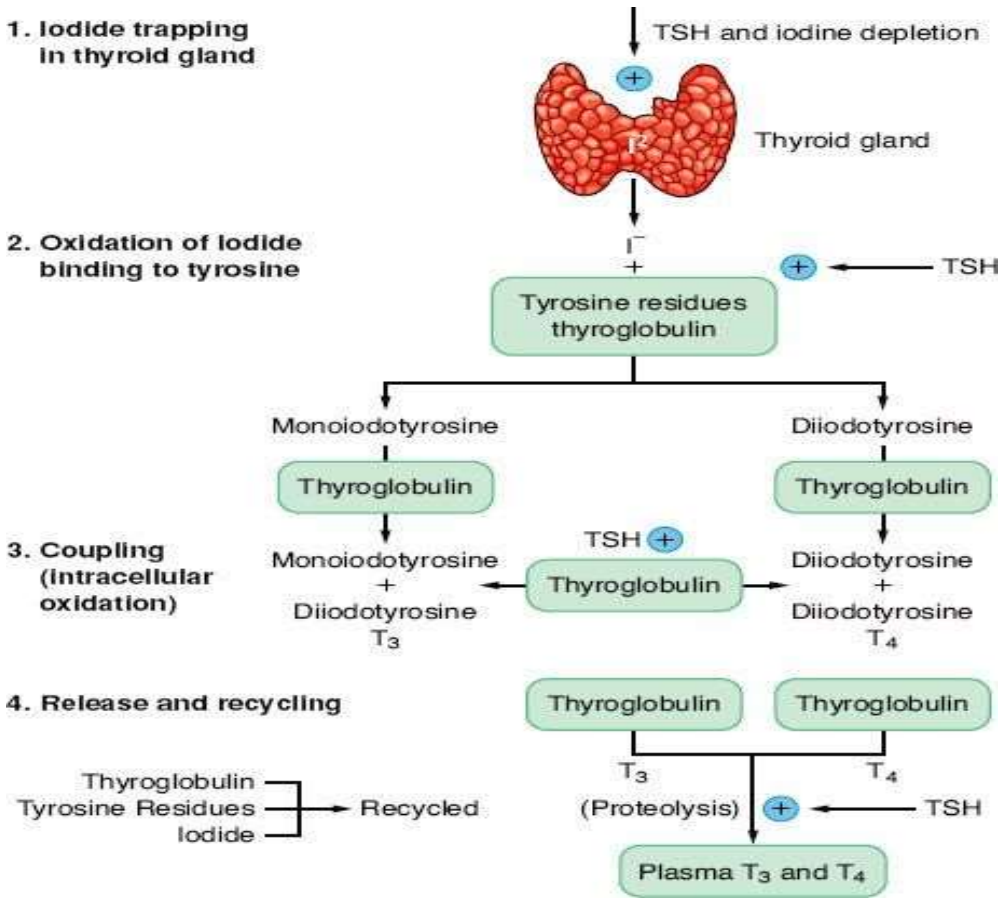
- Inorganic iodine is concentrated within the thyroid follicular cells by a highly efficient, active iodide pump mechanism that removes it from the thyroid blood supply.

- A wide range of anions can compete with iodide for the carrier system and can thereby inhibit the activity of the iodide pump, e.g. ***perchlorate and thiocyanate***.



Biosynthesis and release of Thyroid Hormones

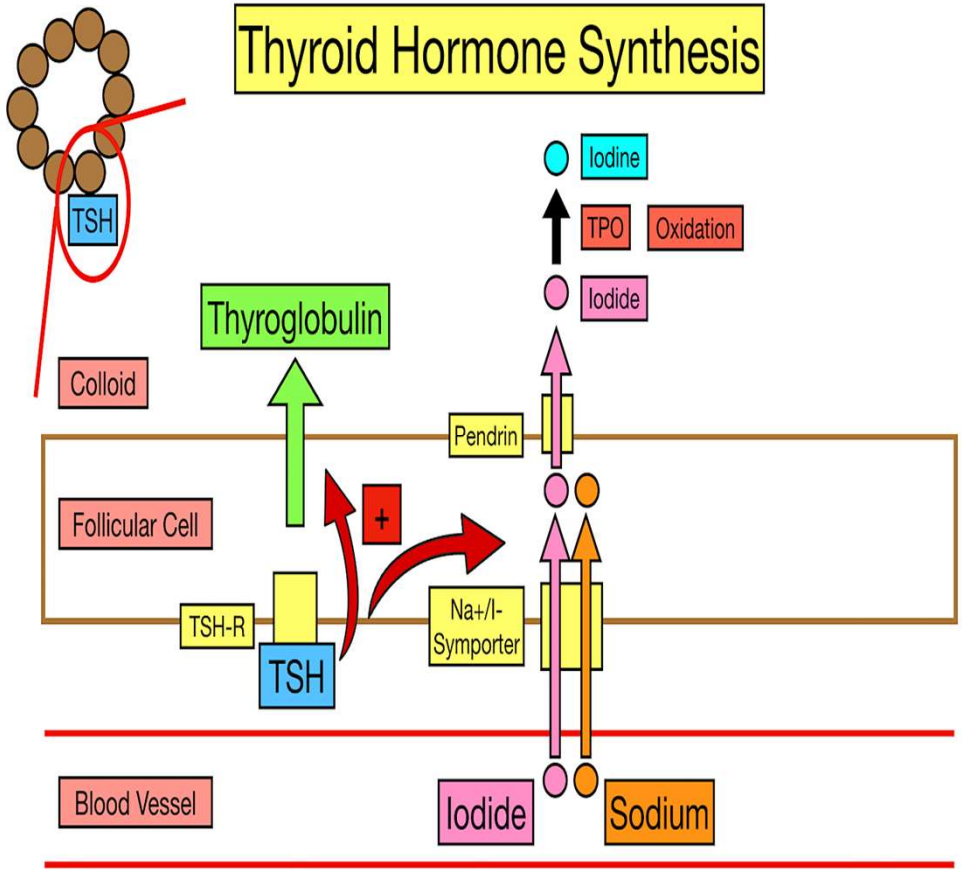
- Hormones are produced in the colloid when atoms of the iodine attach to a glycoprotein called thyroglobulin, which is secreted into the colloid by the follicle cells. The following steps outline the hormones' assembly:



Biosynthesis and release of Thyroid Hormones

A. Trapping iodine

1. The binding of TSH to receptors on thyroid follicle cells prompts the active transport of iodide ions (I^-) from the bloodstream into the cytosol, resulting in a much higher concentration of iodide in the follicular cells than in the blood.
2. Iodide enters the lumen of the follicle cells lining the colloid, where it is oxidized to form iodine (I_2). This iodine then passes through the follicle cell membrane into the colloid.



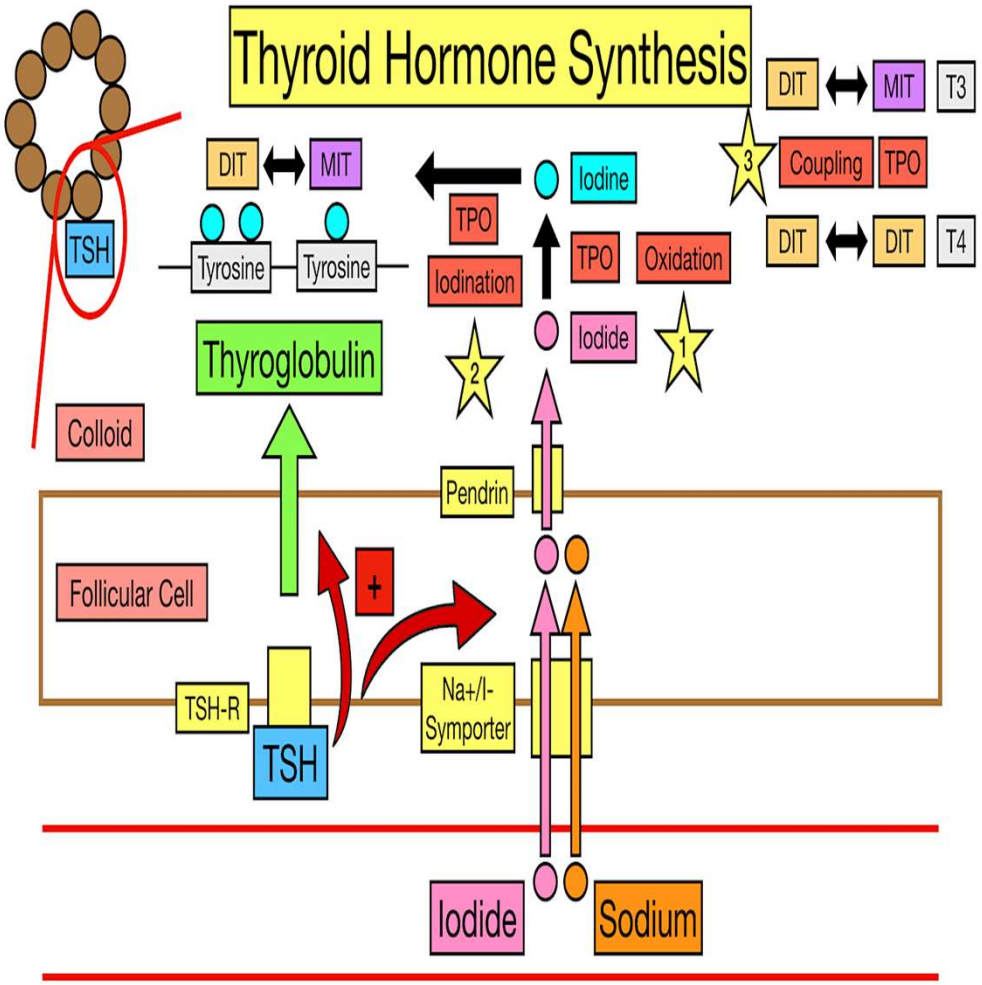
Biosynthesis and release of Thyroid Hormones

B. Iodine oxidation

In the colloid, peroxidase enzymes (TPO) link the iodine to the tyrosine amino acids in thyroglobulin to produce two intermediaries: the 3—and 5-ring positions of the tyrosine molecules form **mono-iodotyrosine (MIT)** and **diiodotyrosine (DIT),** respectively.

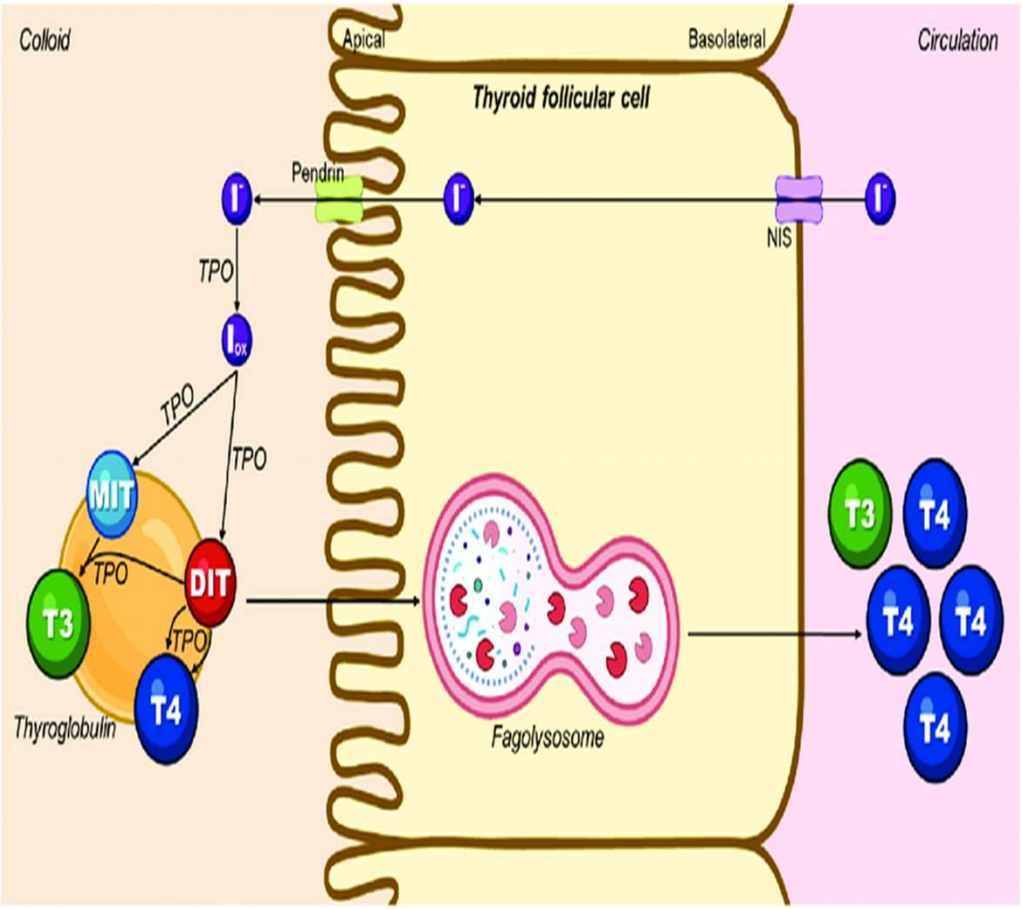
C. Coupling

When intermediaries are linked by covalent bonds, they form triiodothyronine (T3), a thyroid hormone with three iodine atoms. More commonly, two copies of the second intermediary bond together to create tetraiodothyronine (T4), which contains four iodine atoms.



D. Release and recycle

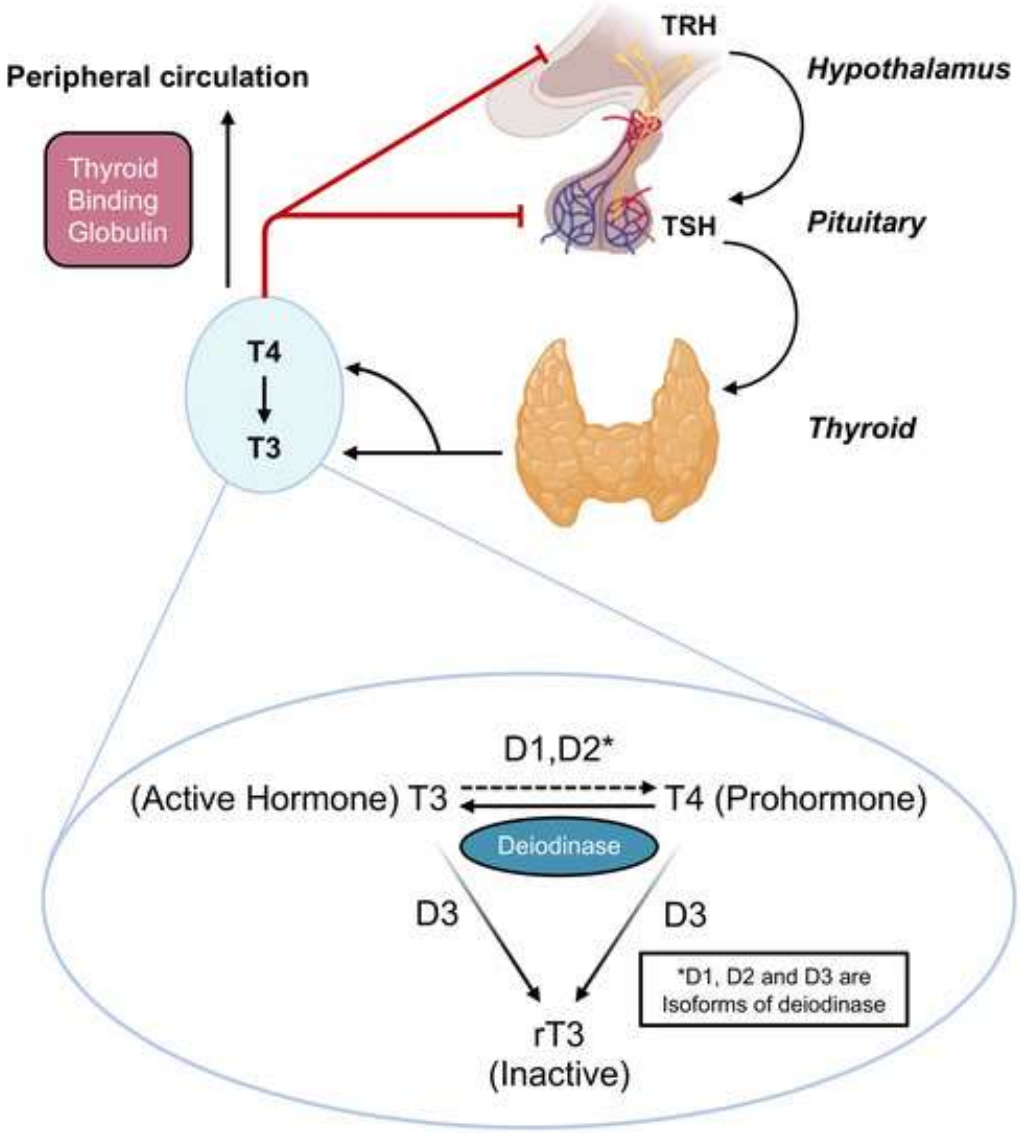
- These hormones remain in the colloid center of the thyroid follicles until TSH stimulates endocytosis of colloid back into the follicle cells. There, lysosomal enzymes break apart the thyroglobulin colloid, releasing free T_3 and T_4, which diffuse across the follicle cell membrane and enter the bloodstream.



- Thyroxine (T_4) accounts for roughly 90 percent of all thyroid secretions, and triiodothyronine (T_3) is secreted in small amounts.

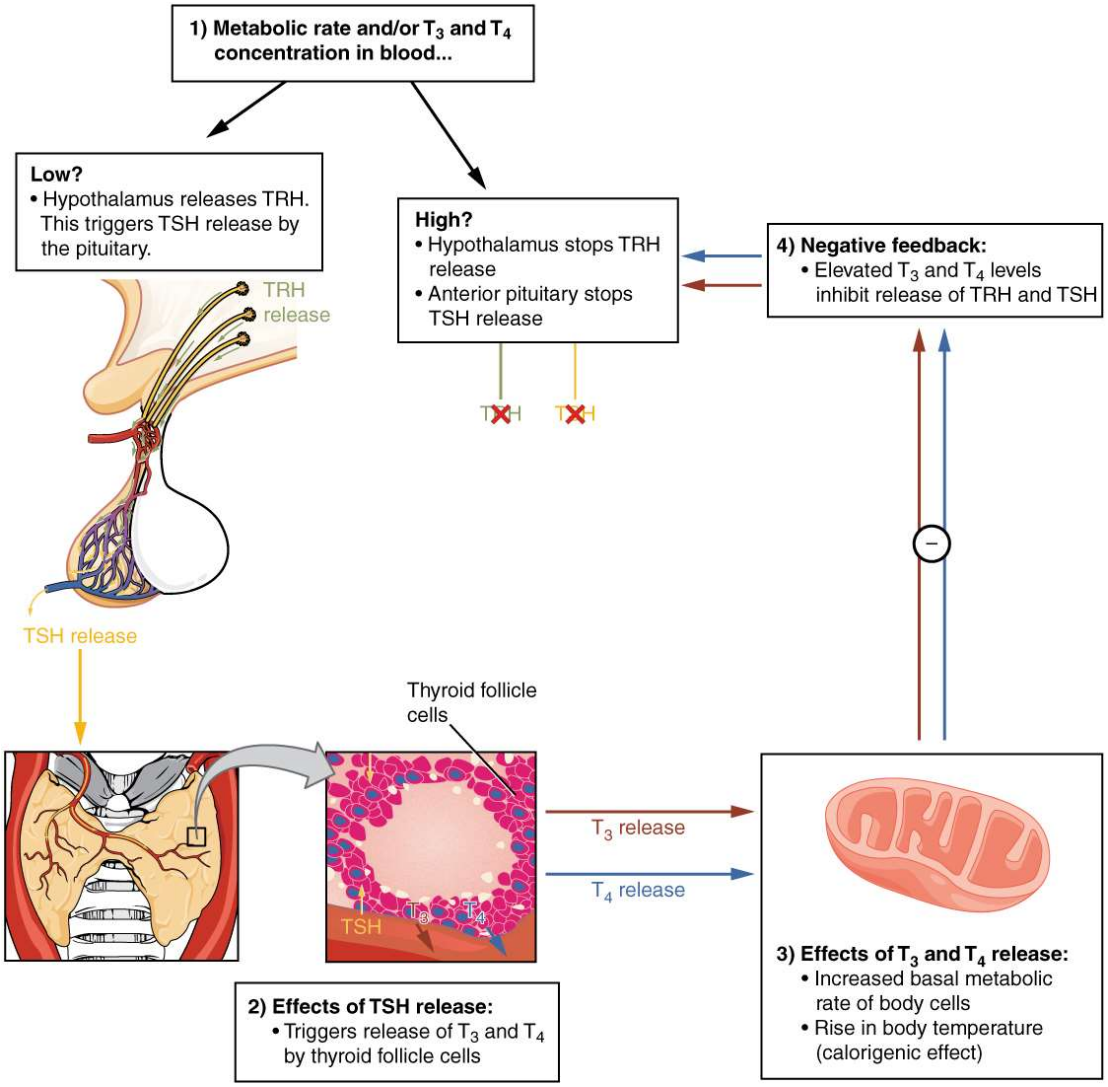
- Roughly 75% of the T_4 and 70% of the T_3 molecules entering the circulation become attached to transport proteins called **thyroid-binding globulins**, or **TBGs**.

- Most of the rest (25%) of the T_4 and T_3 in the circulation is attached to **thyroid-binding prealbumin** or TBPA.



Regulation of Thyroid Hormone Synthesis

- The release of T_3 and T_4 from the thyroid gland is regulated by thyroid-stimulating hormone (TSH).
- Low blood levels of T_3 and T_4 stimulate the release of thyrotropin-releasing hormone (TRH) from the hypothalamus, which triggers the secretion of TSH from the anterior pituitary.
- In turn, TSH stimulates the thyroid gland to secrete T_3 and T_4. The levels of TRH, TSH, T_3, and T_4 are regulated by a negative feedback system in which increasing levels of T_3 and T_4 decrease the production and secretion of TSH.

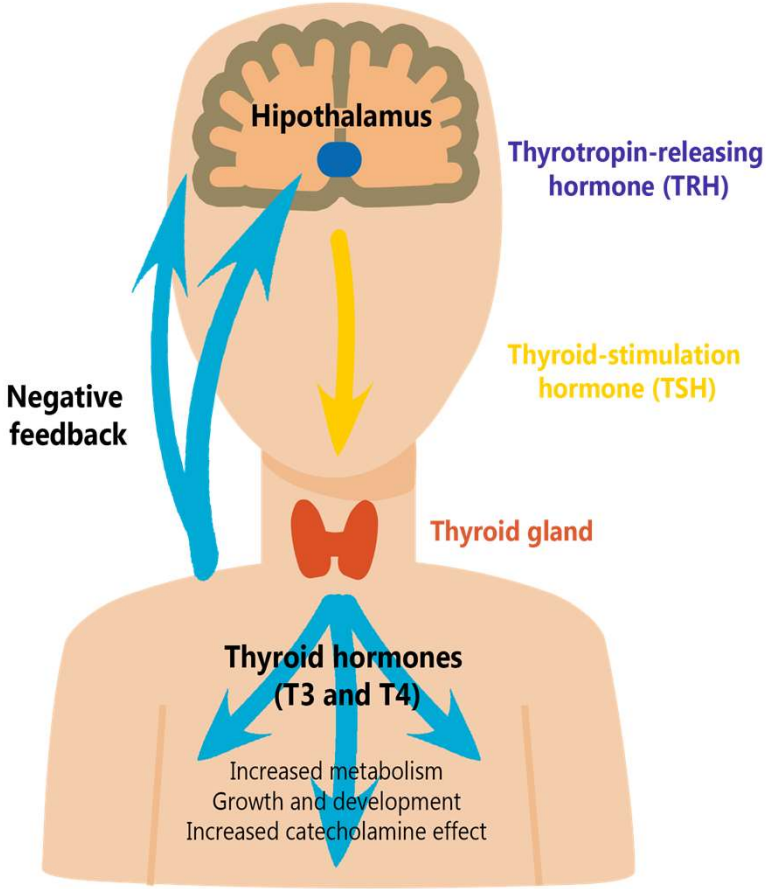


Thyroid hormone Function

- Thyroid hormones affect almost every cell in the body because they readily cross cell membranes. Inside a cell, they bind to (1) receptors in the nucleus, (2) receptors on the surfaces of mitochondria, and (3) receptors in the cytoplasm.

- Thyroid hormones bound to cytoplasmic receptors are essentially held in storage.

- If intracellular levels of thyroid hormones decline, the bound thyroid hormones are released into the cytoplasm.



Actions of thyroid hormones:

T_3 and T_4 have the following effects:

1- General metabolism: Thyroid hormones increase the metabolic rate and O_2 consumption of all tissues of the body except the adult brain, lungs, lymphoid tissues, retina and anterior pituitary gland.

2- Protein metabolism: Normal levels of thyroid hormones stimulate protein synthesis (anabolic effect) whilst excess Thyroid hormones secretions cause protein catabolism (breakdown of protein molecules).

3- Carbohydrate metabolism: Thyroid hormones stimulate glucose uptake and utilization by tissues, insulin hormone secretion, liver glycogenolysis and intestinal absorption of glucose.

4- Lipid metabolism: Thyroid hormones decrease the level of lipid and cholesterol in the blood (i.e. increased cholesterol catabolism than lipogenesis)

Thyroid hormone and fat metabolism

- ***Fat metabolism*** **is generally stimulated**, but **lipolysis is favored**, along with an increased oxidation of free fatty acids.

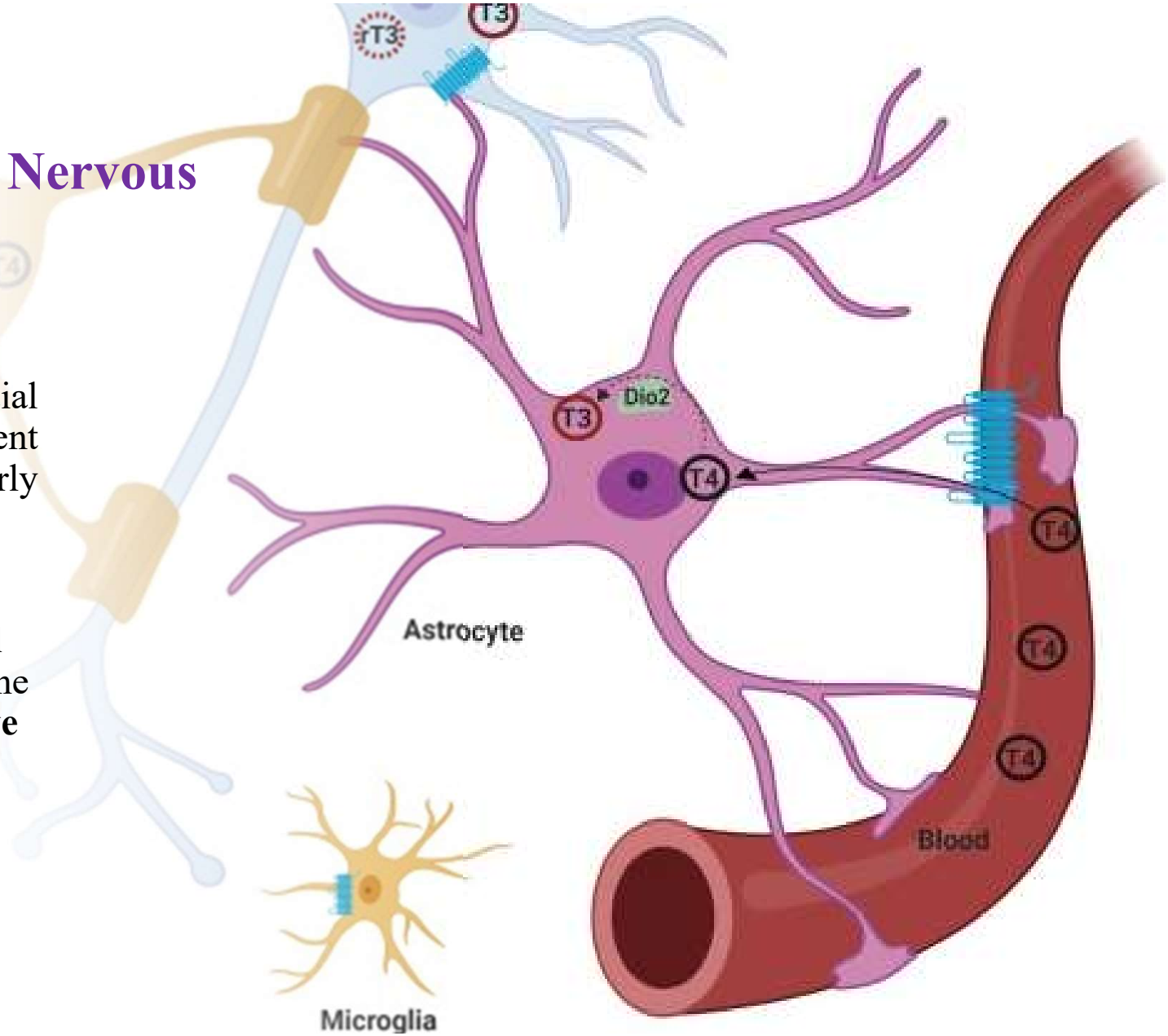
- Part of this lipolytic effect is due to potentiation of catecholamine activity on adipose tissue (a β-adrenergic receptor effect).

- **Plasma cholesterol is lowered by thyroid hormones**, through an **indirect facilitation of liver cholesterol uptake** from the blood (increased synthesis of *low-density lipoprotein (LDL) receptors* in the liver cell membranes).

Maturation of the Central Nervous System (CNS)

- Thyroid hormones are essential for normal CNS development during late fetal and early postnatal life.

- The optimal growth of **cortical and cerebellar neurons**, and the **adequate myelination of nerve fibers** is vitally dependent on their presence.



Cretinism

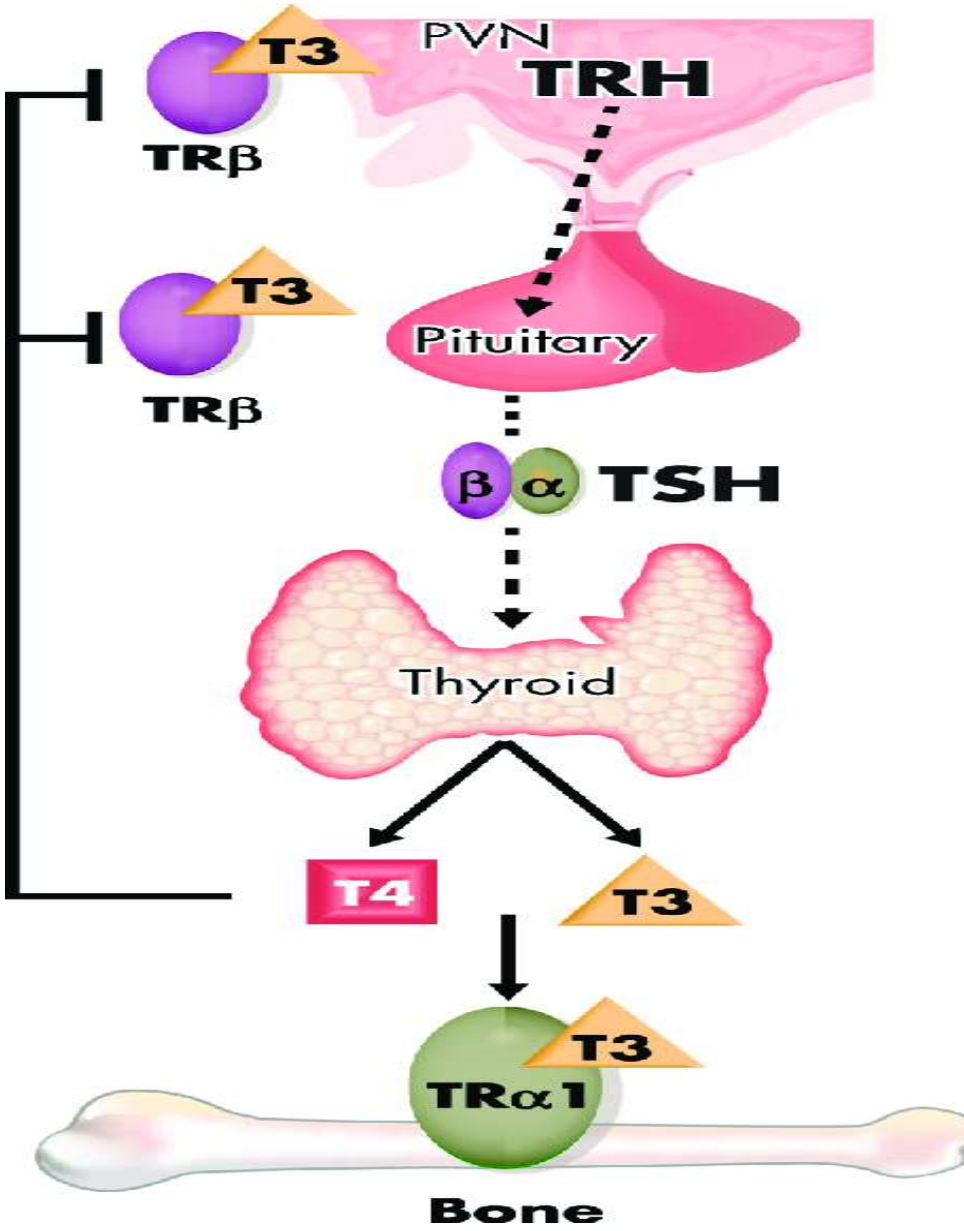
- Their absence or deficiency *in utero* or at birth, if not diagnosed early and promptly treated with thyroid hormone replacement, invariably causes irreversible mental retardation ***(cretinism)***.

Skeletal Growth and Maturation

- The actions of thyroid hormone are generally synergistic with those of *growth hormone (GH)*.

- Thyroid hormone is thus essential for normal bone growth and maturation, and the eventual development of normal adult stature.

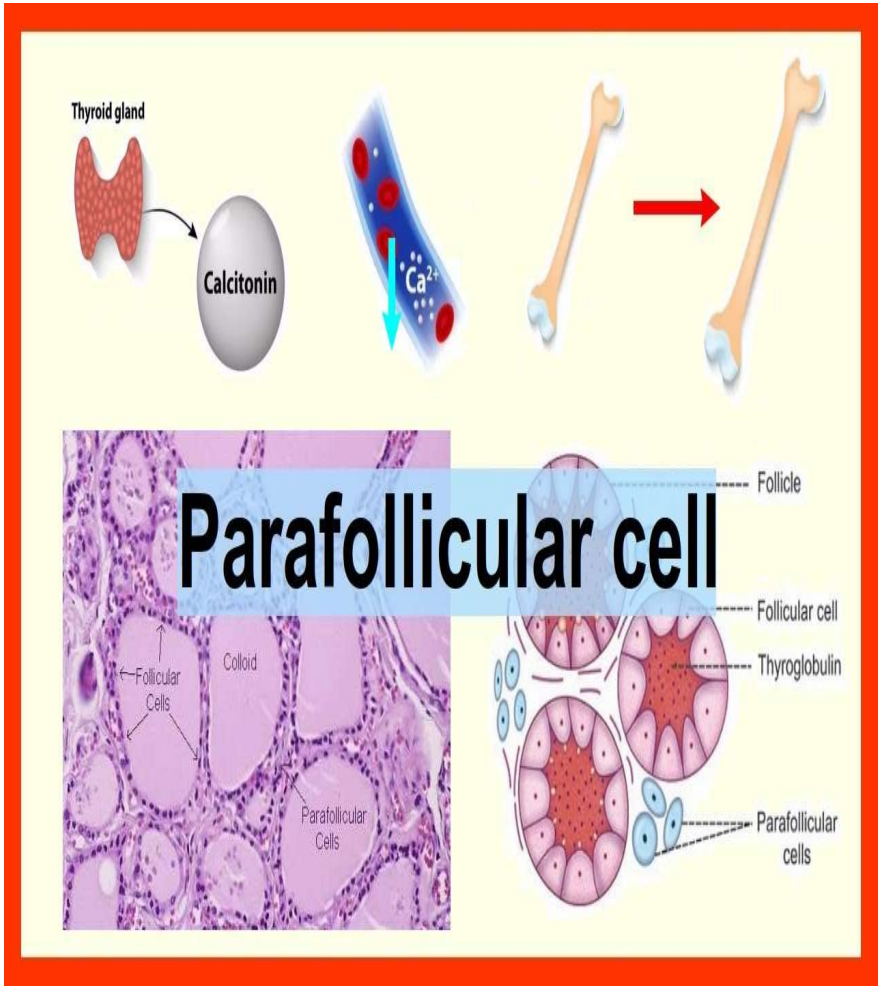
- Normal amounts of thyroid hormone are also necessary for proper functioning of the nervous and cardiovascular systems, the gastrointestinal tract, as well as for regular development of the teeth, skin and hair follicles.



Thyroid Parafollicular (C) cells and Calcitonin

- The blood calcium level is regulated in part by **calcitonin,** a hormone secreted by the parafollicular cells (C cells) of thyroid gland when the blood calcium level rises.

- Calcium plays a significant role in both nervous conduction and muscle contraction. It is also necessary to blood clotting.



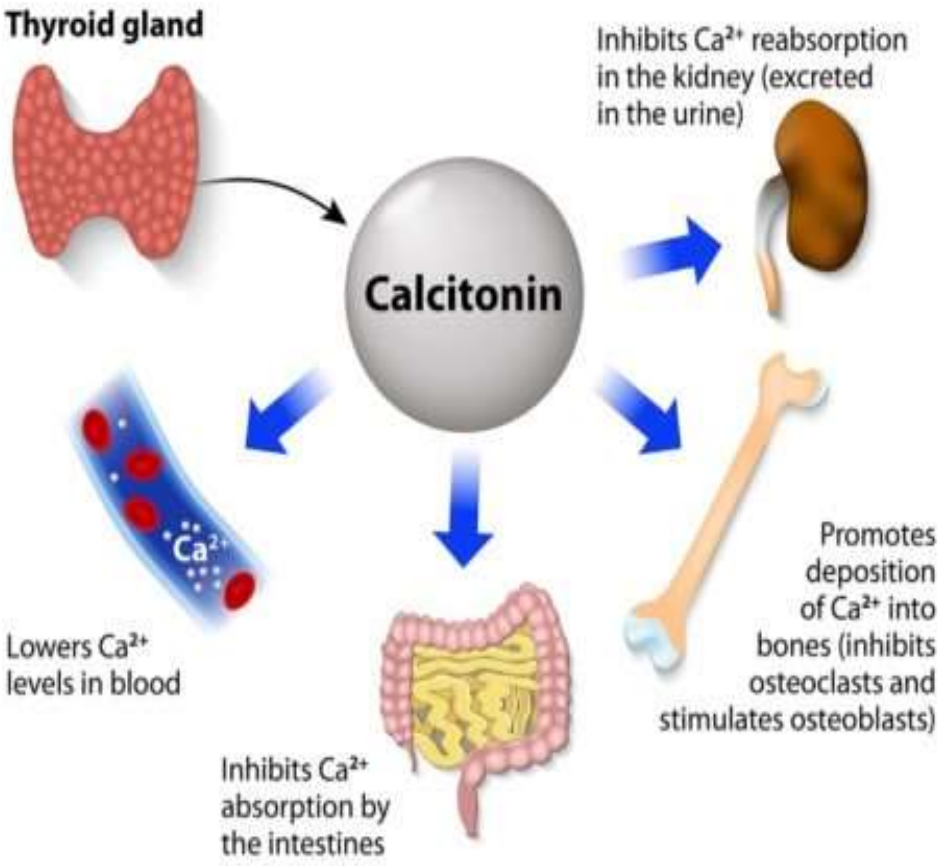
Calcitonin has the following effects:

1-**Stimulates** **osteoblasts**, leading to increased bone deposition and the removal of calcium ions from body fluids.

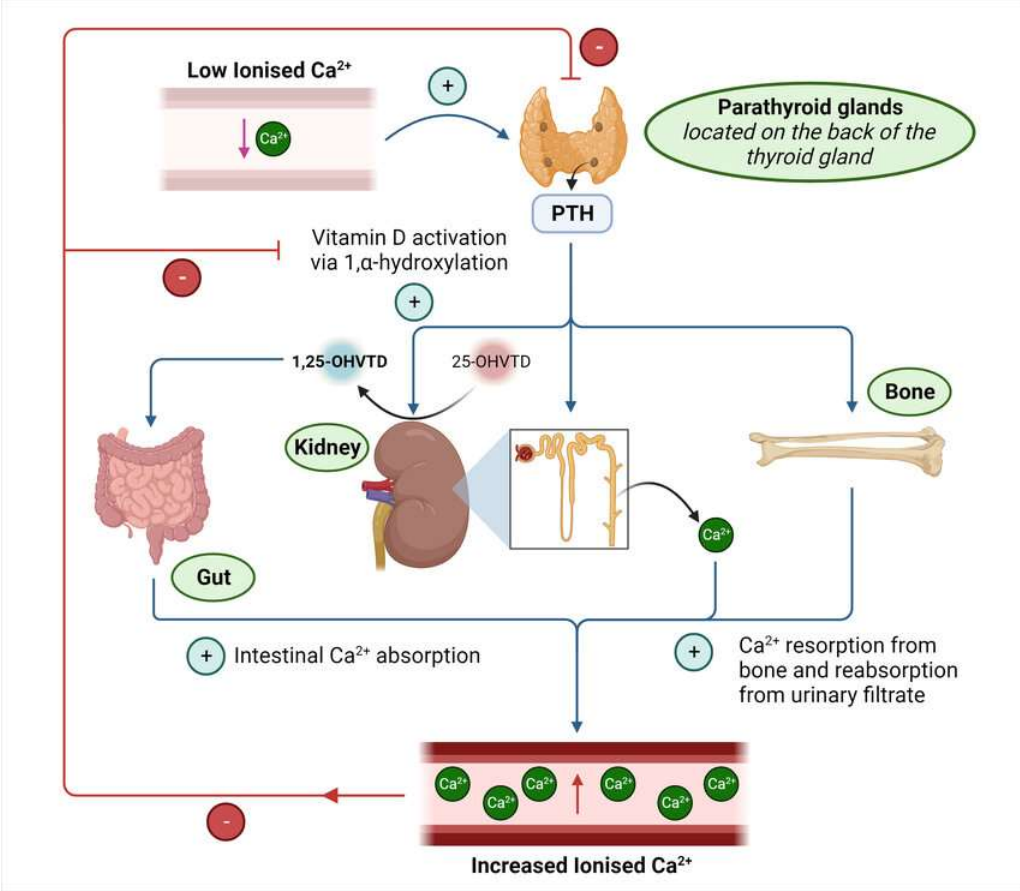
2-**Inhibits** **osteoclasts**, slowing the rate of calcium ion release from bone.

3-**Stimulates calcium ion excretion** at the **kidneys**.

4-Inhibits secretion of the **antagonistic** hormone, **parathyroid** hormone.

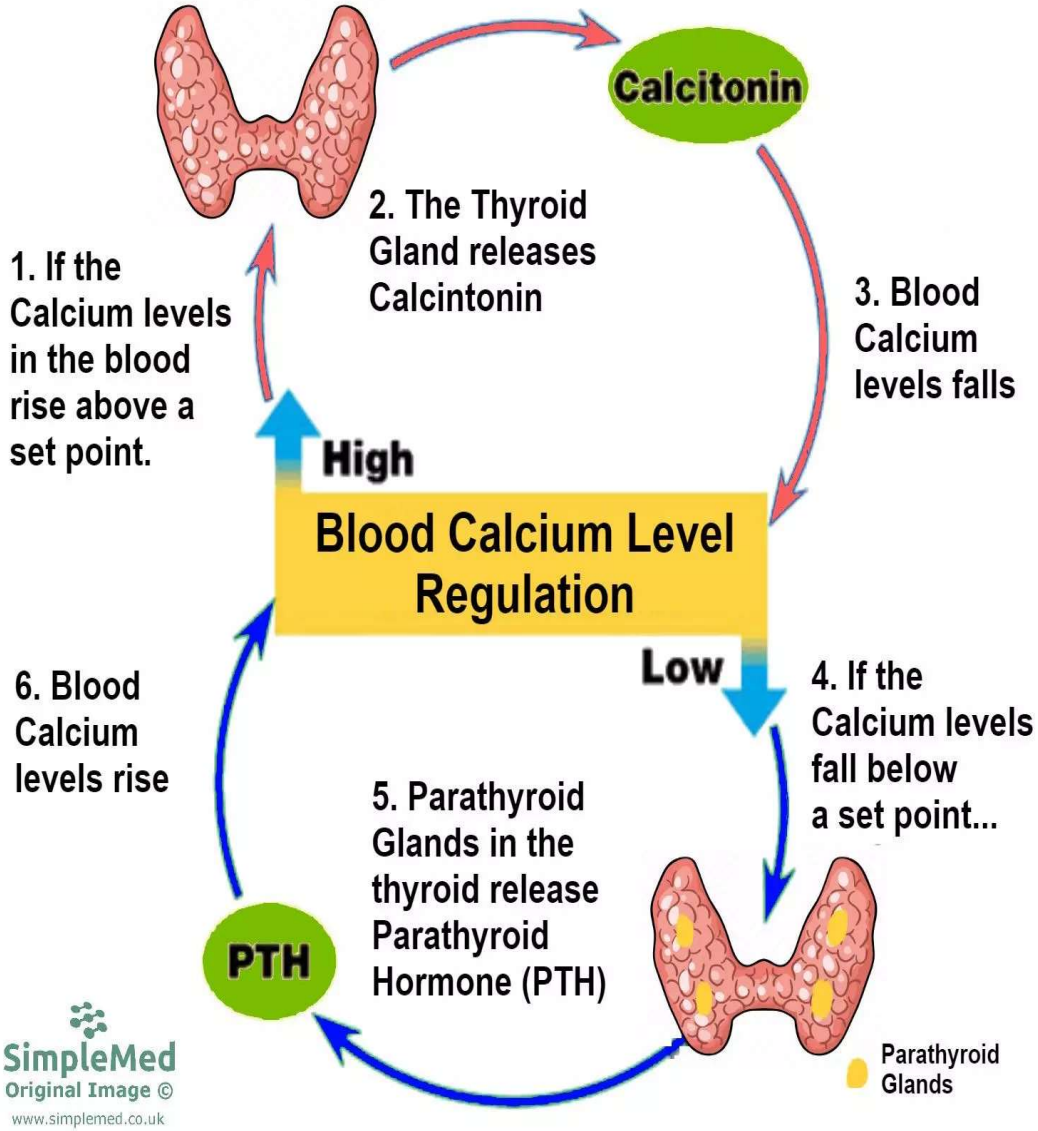


Calcium hemostasis



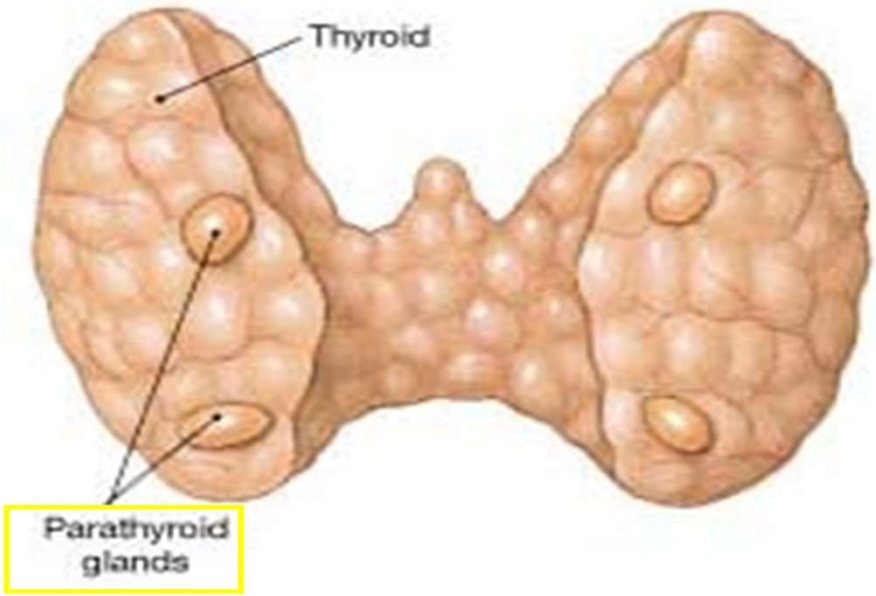
- When the blood calcium decreases to normal, the release of calcitonin by the thyroid is inhibited by negative feed back mechanism.

- On the other hand, low levels of Ca stimulates the release of **parathyroid hormone (PTH)** by the parathyroid glands.



Parathyroid glands

- The **parathyroid glands** are 4 small glands embedded in the posterior surface of the thyroid gland.
- It is secreted parathyroid hormone (PTH) that is released when blood calcium level is low.



(a) Thyroid gland, posterior view

- PTH has the following effects:

1. **In bones, it stimulates osteoclasts**, accelerating mineral turnover and the release of calcium ions from bone.
2. **In bones, it inhibits osteoblasts**, reducing the rate of calcium deposition in bone.
3. **In GIT, it promotes** the **intestinal absorption** of calcium and phosphate ions.
4. **In Kidneys, it reduces urinary excretion** of calcium ions. It also **stimulates the formation and secretion of *calcitriol* in the kidneys, which is a** third ca^{2+}-regulating hormone.

These effects bring the blood calcium level back to the normal range so that the parathyroid glands no longer secrete PTH.

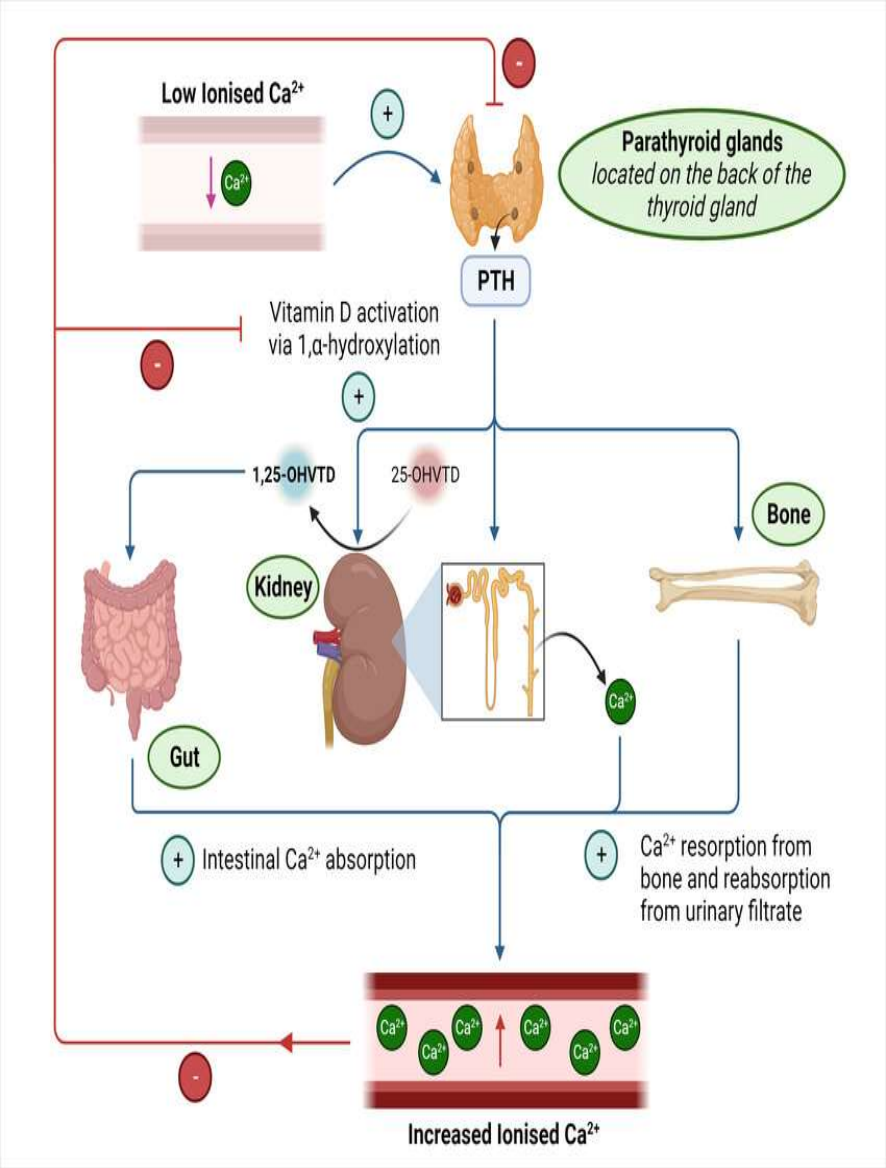
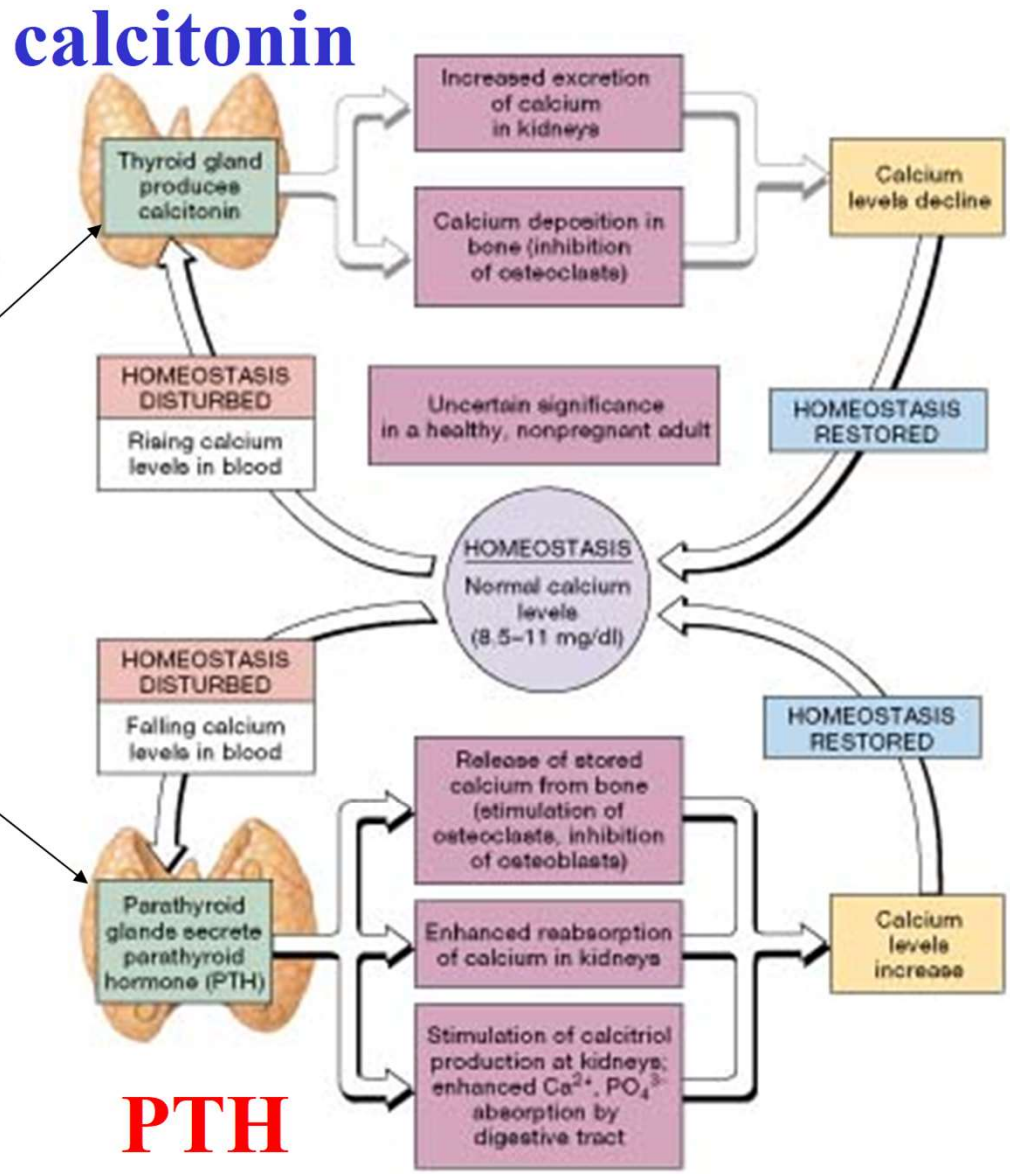


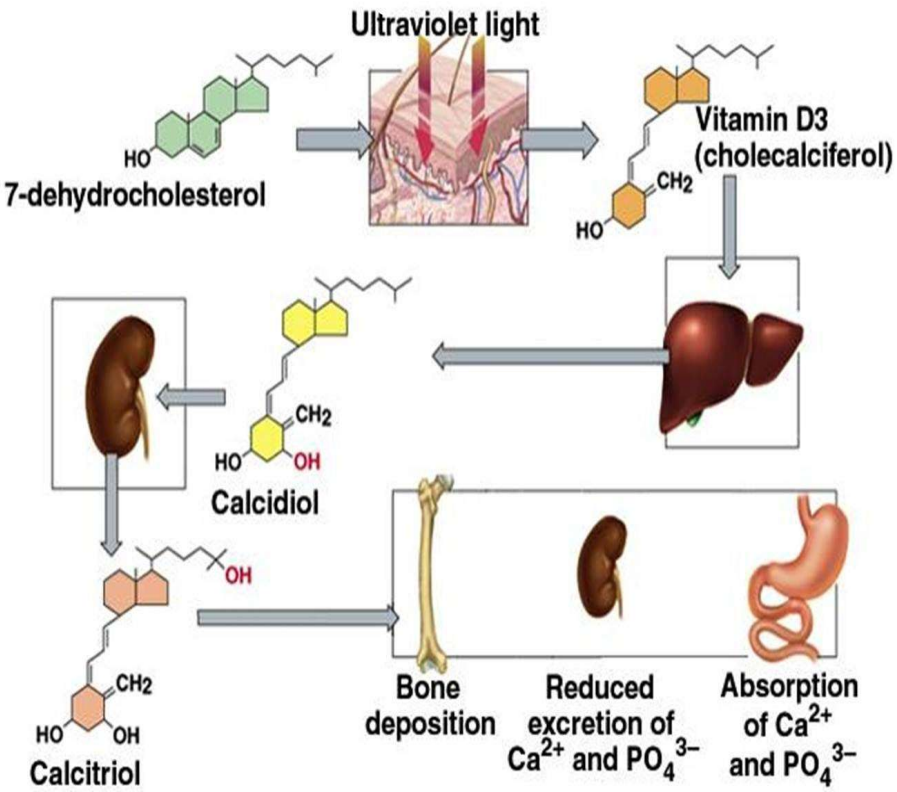
FIGURE 18-15
The Homeostatic Regulation of Calcium Ion Concentrations

Antagonistic hormones



Calcitriol

- Calcitriol is a **steroid hormone** secreted by the **kidney** in response to the presence of parathyroid hormone (PTH).
- Calcitriol synthesis is dependent on the availability of a related steroid, ***cholecalciferol* (vitamin D_3)**, which may be synthesized in the skin or absorbed from the diet.
- Cholecalciferol from either source is absorbed from the bloodstream by the liver and converted to an intermediary product.
- This intermediary is released into the circulation and absorbed by the kidneys for conversion to calcitriol.



- The best-known function of calcitriol is **stimulation of calcium and phosphate ions absorption along the digestive tract**.

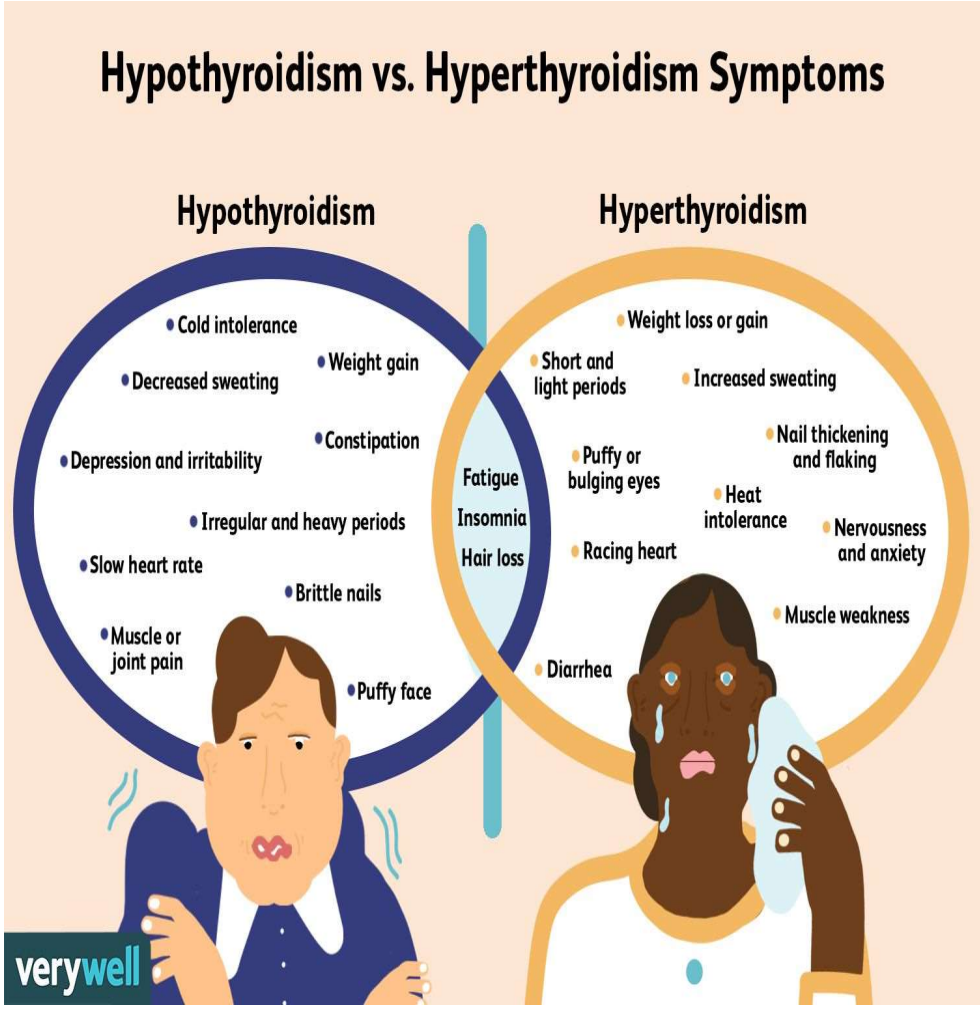
- The effects of PTH on calcium ion absorption result at least partially via stimulation of calcitriol release. Hence, calcitriol is considered as **PTH agonist**.

Clinical Disorders

- Apart from *diabetes mellitus,* thyroid dysfunction is one of the most common endocrine disorders encountered in adults.

- Females are generally more susceptible to thyroid disorders than males.

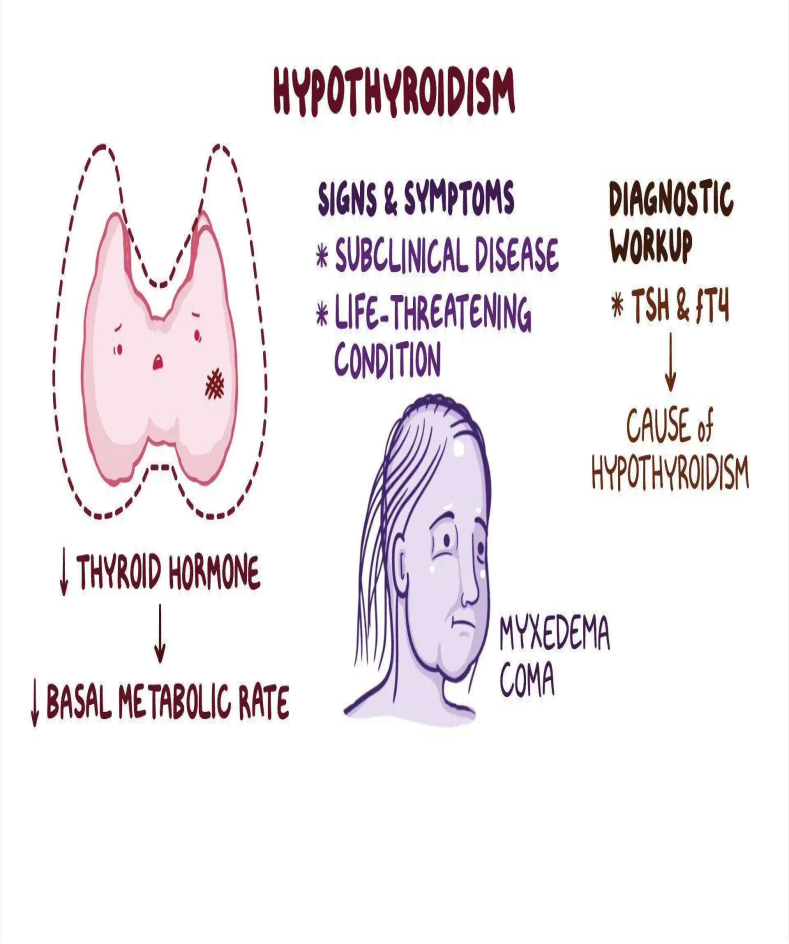
- The two most common thyroid diseases are inadequate thyroid function, ***hypothyroidism***, and excessive thyroid function, ***hyperthyroidism (thyrotoxicosis)***.



ADULT HYPOTHYROIDISM

- Hypothyroidism can develop as a result of a functional defect in the thyroid itself ***(primary hypothyroidism)*** or as a consequence of reduced stimulatory input from the pituitary TSH ***(secondary hypothyroidism)***.

- Primary **hypothyroidism** is most commonly caused by an **autoimmune destruction** of the thyroid gland, leading to a deficiency in the amounts of circulating thyroid hormones ***(Hashimoto's thyroiditis)*.**

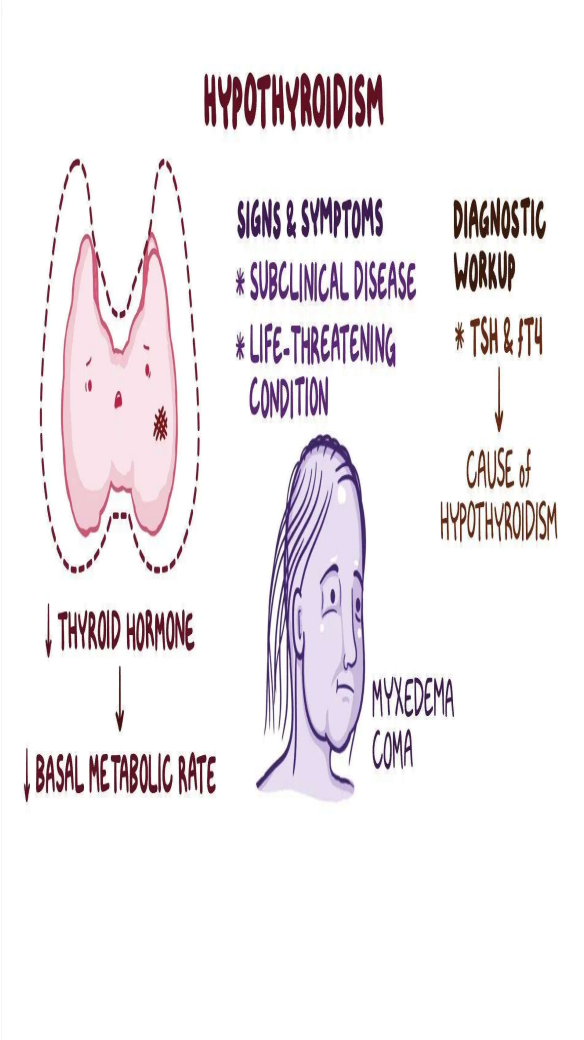


ADULT HYPOTHYROIDISM

- It is characterized by the appearance of **destructive serum antibodies** to thyroglobulin and to cytoplasmic components of the thyroid follicular cells (microsomal peroxidase enzyme), and a progressive infiltration of the gland by lymphocytes and plasma cells.

- A painful diffuse swelling of the gland ***(goiter)*** may be present in the early stages of the disease.

- ***Goiter*** is the name given to any **enlargement of the thyroid** that may be associated with decreased, normal or increased thyroid activity.

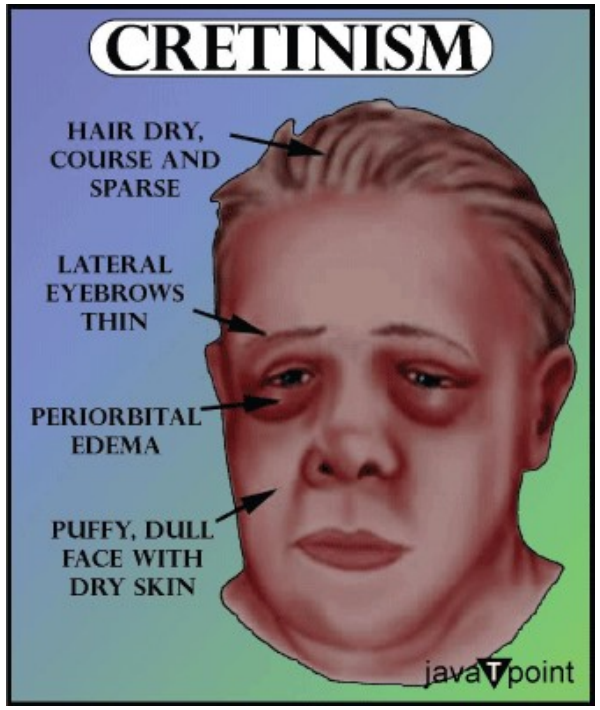


CHILDHOOD (CONGENITAL) HYPOTHYROIDISM

- Uncorrected hypothyroidism at birth (usually due to failure of proper development of the thyroid) leads to ***cretinism;*** infant subjects are dwarfed, with a protruding tongue and abdomen, mentally retarded, and have coarse scanty hair and dry, yellowish skin.

- Development of primary hypothyroidism in early childhood ***(juvenile hypothyroidism)*** is usually characterized by stunted growth, delayed sexual development, and mental slowness, usually reflected by poor performance at school.

- The replacement dose of thyroxine (25–200 μg daily) needs to be carefully adjusted to ensure normal growth.

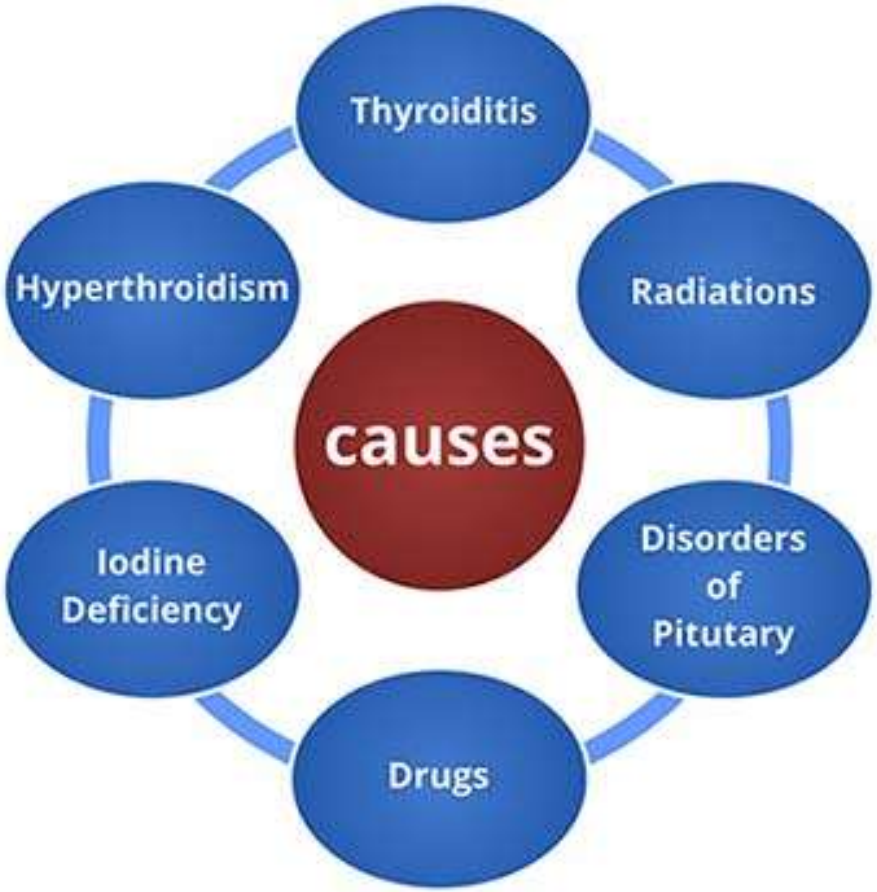


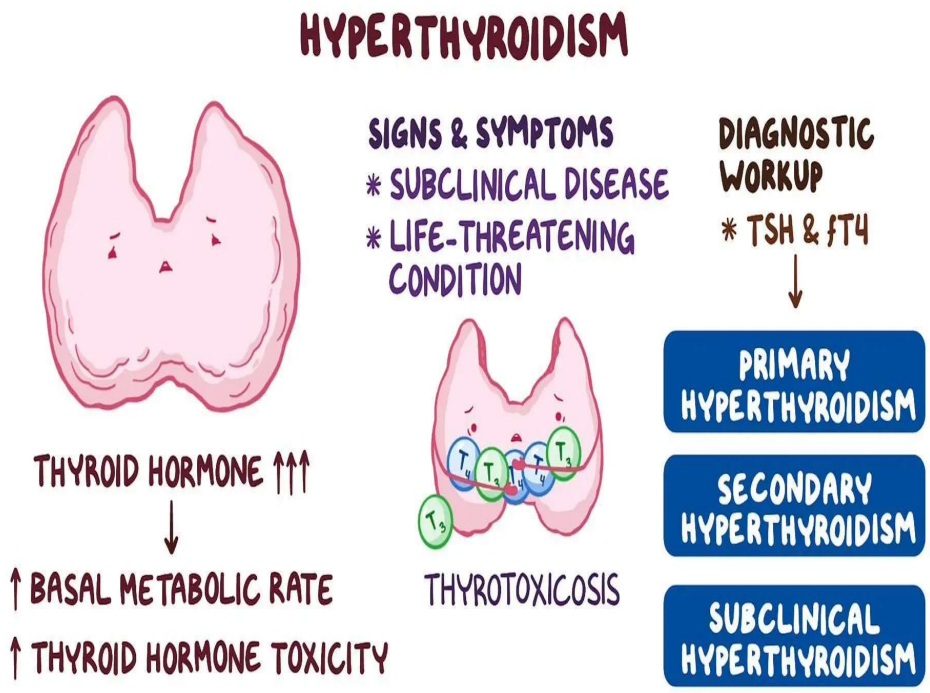
Some other important causes of hypothyroidism include:

1. ***Surgical removal (partial thyroidectomy)* or *radioactive iodine (^{131}I)* treatment of thyroid tissue** for the relief of *hyper*thyroidism.

2. ***Inadequate TSH production*** due to pituitary or hypothalamic disease; a goiter or myxoedema (severe hypothyroidism) does not normally accompany this rare secondary condition.

3. ***Iodine deficiency:*** Endemic goiter can result from low iodide intake, often found in seafood. While rare in countries with iodized salt, it can still occur in remote or mountainous regions.



Hyperthyroidism

- Hyperthyroidism, also called **overactive thyroid**, is a condition in which the thyroid produces and releases high levels of thyroid hormone.
- Hyperthyroidism speeds up metabolism and heart rate which can affect several aspects of health.
- Hyperthyroidism is relatively rare.